

Unit I :: Frames of Reference

 **PHY01MX01R04: Mechanics and Properties of Matter** 

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Syllabus

Unit I: Frames of Reference

Review of Newton's laws of motion; inertial frames of reference, Galilean transformations, Galilean invariance; non-inertial frames of reference, fictitious (inertial) forces – centrifugal force, Coriolis force; transformation equations in inclined and rotating frames.

1 Review of Newton's laws of motion

Before delving into a review of Newton's laws of motion, we need to be clear of the definitions of some important terms used in mechanics:

- *Particle*: an idealized object that has mass and position but occupies zero volume.
- *Event*: a specific occurrence that happens at a unique place in space and a specific instant in time.
- *Observer*: a person or an equipment which can locate, record, measure and interpret an event.
- *Frame of reference*: a coordinate system combined with a clock, used by an observer to measure the position and motion of particles.
- *Rest*: a particle is said to be at rest when its position does not change relative to a chosen frame of reference over time.
- *Motion*: the change in position or orientation of an object with respect to a chosen frame of reference over time.
- *Inertia*: the tendency of a body to resist any change in its motion or configuration.
- *Position*: the location of a particle in space with respect to a fixed reference point known as the origin. If (x, y, z) are the coordinates of a particle in Cartesian space, then its position vector is

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}} \quad (1)$$

- *Displacement*: the change in the position of a particle. If a particle is initially at the position $\mathbf{r}_i = x_i\hat{\mathbf{i}} + y_i\hat{\mathbf{j}} + z_i\hat{\mathbf{k}}$ and then finally moves to the position $\mathbf{r}_f = x_f\hat{\mathbf{i}} + y_f\hat{\mathbf{j}} + z_f\hat{\mathbf{k}}$, its displacement vector is

$$\Delta\mathbf{r} = \mathbf{r}_f - \mathbf{r}_i = (x_f - x_i)\hat{\mathbf{i}} + (y_f - y_i)\hat{\mathbf{j}} + (z_f - z_i)\hat{\mathbf{k}} \quad (2)$$

If the displacement is infinitesimally small, then we can write

$$d\mathbf{r} = \hat{\mathbf{i}}dx + \hat{\mathbf{j}}dy + \hat{\mathbf{k}}dz \quad (3)$$

- *Velocity*: the rate of change of position of a particle. If the position of a particle varies with time, then its instantaneous velocity is the first derivative of its position with respect to time.

$$\mathbf{v} = \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \quad (4)$$

If the positions of a particle at two instants of time are $\mathbf{r}(t_1) = \mathbf{r}_1$ and $\mathbf{r}(t_2) = \mathbf{r}_2$, then its displacement is obtained by the integral

$$\begin{aligned} \int_{\mathbf{r}_1}^{\mathbf{r}_2} d\mathbf{r} &= \int_{t_1}^{t_2} \mathbf{v} dt \\ \Rightarrow \mathbf{r}_2 - \mathbf{r}_1 &= \int_{t_1}^{t_2} \mathbf{v} dt \end{aligned} \quad (5)$$

- *Acceleration*: the rate of change of velocity of a particle. If the velocity of a particle varies with time, then its instantaneous acceleration is the first derivative of its velocity with respect to time, or the second derivative of its position with respect to time.

$$\mathbf{a} = \dot{\mathbf{v}} = \frac{d\mathbf{v}}{dt} \quad (6)$$

$$\text{or } \mathbf{a} = \ddot{\mathbf{r}} = \frac{d^2\mathbf{r}}{dt^2} \quad (7)$$

If the velocities of a particle at two instants of time are $\mathbf{v}(t_1) = \mathbf{v}_1$ and $\mathbf{v}(t_2) = \mathbf{v}_2$, then the change in its velocity is obtained by the integral

$$\begin{aligned} \int_{\mathbf{v}_1}^{\mathbf{v}_2} d\mathbf{v} &= \int_{t_1}^{t_2} \mathbf{a} dt \\ \Rightarrow \mathbf{v}_2 - \mathbf{v}_1 &= \int_{t_1}^{t_2} \mathbf{a} dt \end{aligned} \quad (8)$$

1.1 Kinematics relations

The kinematic relations are a set of formulae in mechanics that describe the motion of objects moving with constant acceleration, without considering the forces that cause the motion.

1.1.1 Velocity-time relation

Consider the motion of a particle moving with a constant acceleration $\mathbf{a}$. Let its initial velocity be $\mathbf{v}_1 = \mathbf{v}_0$ at time $t_1 = 0$, and its final velocity be $\mathbf{v}_2 = \mathbf{v}(t)$ at some arbitrary time t . Then from equation (8), we have

$$\begin{aligned} \mathbf{v}(t) - \mathbf{v}_0 &= \mathbf{a} \int_0^t dt \\ \Rightarrow \mathbf{v}(t) - \mathbf{v}_0 &= \mathbf{a}(t - 0) \\ \Rightarrow \boxed{\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t} \end{aligned} \quad (9)$$

which is the *velocity-time relation*, or the *first kinematic relation*.

1.1.2 Position-time relation

Consider the motion of a particle moving with a constant acceleration $\mathbf{a}$. Let its initial position be $\mathbf{r}_1 = \mathbf{r}_0$ at time $t_1 = 0$, and its final position be $\mathbf{r}_2 = \mathbf{r}(t)$ at some arbitrary time t . Then from

equation (5), we have

$$\begin{aligned}
 \mathbf{r}(t) - \mathbf{r}_0 &= \int_0^t \mathbf{v} dt \\
 &= \int_0^t (\mathbf{v}_0 + \mathbf{a}t) dt \quad [\text{using eqn. (9)}] \\
 &= \mathbf{v}_0 \int_0^t dt + \mathbf{a} \int_0^t t dt \\
 &= \mathbf{v}_0(t - 0) + \mathbf{a} \frac{t^2}{2} \Big|_0^t \\
 \Rightarrow \boxed{\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2} & \quad (10)
 \end{aligned}$$

which is the *position-time relation*, or the *second kinematic relation*.

1.1.3 Position-velocity relation

We know that

$$\mathbf{a} = \frac{d\mathbf{v}}{dt}$$

Taking the dot product of the above with an infinitesimal displacement $d\mathbf{r}$, we have

$$\begin{aligned}
 \mathbf{a} \cdot d\mathbf{r} &= \frac{d\mathbf{v}}{dt} \cdot d\mathbf{r} \\
 \Rightarrow \mathbf{a} \cdot d\mathbf{r} &= \frac{d\mathbf{r}}{dt} \cdot d\mathbf{v} \\
 \Rightarrow \mathbf{a} \cdot d\mathbf{r} &= \mathbf{v} \cdot d\mathbf{v} \quad (11)
 \end{aligned}$$

Now, using the product rule for differentiation, we have

$$\begin{aligned}
 d(\mathbf{v} \cdot \mathbf{v}) &= d\mathbf{v} \cdot \mathbf{v} + \mathbf{v} \cdot d\mathbf{v} \\
 \Rightarrow 2\mathbf{v} \cdot d\mathbf{v} &= d(\mathbf{v} \cdot \mathbf{v}) \\
 \Rightarrow \mathbf{v} \cdot d\mathbf{v} &= \frac{1}{2} d(\mathbf{v} \cdot \mathbf{v})
 \end{aligned}$$

Then equation (11) becomes

$$\mathbf{a} \cdot d\mathbf{r} = \frac{1}{2} d(\mathbf{v} \cdot \mathbf{v}) \quad (12)$$

Suppose we have a particle moving under constant acceleration $\mathbf{a}$. If at its initial and final positions $\mathbf{r}_0$ and $\mathbf{r}$, it has initial and final velocities $\mathbf{v}_0$ and $\mathbf{v}$, then we can integrate equation (eqn:adr-

v dv-02) as

$$\begin{aligned}
 \mathbf{a} \cdot \int_{\mathbf{r}_0}^{\mathbf{r}} d\mathbf{r} &= \frac{1}{2} \int_{\mathbf{v}_0}^{\mathbf{v}} d(\mathbf{v} \cdot \mathbf{v}) \\
 \Rightarrow \mathbf{a} \cdot \mathbf{r} \Big|_{\mathbf{r}_0}^{\mathbf{r}} &= \frac{1}{2} \mathbf{v} \cdot \mathbf{v} \Big|_{\mathbf{v}_0}^{\mathbf{v}} \\
 \Rightarrow \mathbf{a} \cdot (\mathbf{r} - \mathbf{r}_0) &= \frac{1}{2} (\mathbf{v} \cdot \mathbf{v} - \mathbf{v}_0 \cdot \mathbf{v}_0) \\
 &= \frac{1}{2} (v^2 - v_0^2) \\
 \Rightarrow \boxed{v^2} &= v_0^2 + 2\mathbf{a} \cdot (\mathbf{r} - \mathbf{r}_0) \quad (13)
 \end{aligned}$$

which is the *position-velocity relation*, or the *third kinematic relation*.

1.2 Newton's laws of motion

1.2.1 Newton's first law of motion

A particle tends to maintain its state of rest or uniform motion unless it is acted upon by an external agency.

- The importance of Newton's first law of motion lies in that it defines force as:

Force is any external agency that brings about a change in the state of rest or uniform motion of a particle.

1.2.2 Newton's second law of motion

The rate of change of momentum of a particle is directly proportional to the force exerted on it, and takes place in the direction of the force.

$$\mathbf{F} \propto \frac{d\mathbf{p}}{dt}$$

This implies that we have

$$\mathbf{F} = k \frac{d\mathbf{p}}{dt}$$

where k is the proportionality constant.

Now, for a particle of mass m and moving with a velocity $\mathbf{v}$, its momentum is defined as $\mathbf{p} = m\mathbf{v}$. Hence, Newton's second law becomes

$$\begin{aligned}
 \mathbf{F} &= k \frac{d}{dt}(m\mathbf{v}) \\
 &= km \frac{d\mathbf{v}}{dt} \quad [\text{assuming } m \text{ to be constant}] \\
 \mathbf{F} &= k m \mathbf{a}
 \end{aligned}$$

The constant k is set to 1 by our choice of the units of force. We define 1 newton to be that amount of force that brings about an acceleration of 1 m/s^2 on a particle of mass 1 kg. Then from the above, we have

$$(1 \text{ N}) = k \times (1 \text{ kg}) \times (1 \text{ m/s}^2) \\ \Rightarrow k = 1$$

Therefore, for a particle *whose mass remains unchanged*, Newton's second law of motion can be written as

$$\boxed{\mathbf{F} = m\mathbf{a}}$$

- The importance of Newton's second law of motion lies in that it measures force as:

If a force of magnitude F brings about an acceleration a on a particle of mass m , then the force is measured to be $F = ma$.

1.2.3 Newton's third law of motion

For every action, there is an equal and opposite reaction, both of which act on different bodies.

$$\mathbf{F}_{12} = -\mathbf{F}_{21}$$

- The importance of Newton's second law of motion lies in that it gives the origin of force as:

If a particle experiences a force of magnitude F , it is because it exerts another force of the same magnitude F but in the opposite direction on some other particle.

1.3 Special cases of Newton's second law of motion

1.3.1 The first law as a special case

1.3.2 The third law as a special case

2 Inertial frames of reference

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3 Non-inertial frames of reference

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